

CIS 371 Web Application Programming

JS|TS Modules



Lecturer: **Dr. Haoyu Li**



Topics

- Why Modules?
 - The problem we are trying to solve
- Solution
- Examples

Multiple Scripts: name conflicts

```
<html>
  <body>
    <script src="one.js"></script>
    <script src="two.js"></script>
    <script>
      let msg = "Here";
    </script>
    <script>
      let msg = "I'm here";
    </script>
    <script>
      console.debug(msg);
    </script>
  </body>
</html>
```

```
// one.js
let msg = "Hello";
```

```
// two.js
let msg = "Hello World";
```

- *JS Error: "msg" is already defined*
- *Variables defined in each <script> block are globally visible*

Solution: use Modules (ES6)

```
<html>
  <body>
    <script type="module">
      let msg = "Here";
    </script>
    <script type="module">
      let msg = "I'm here";
    </script>
    <script type="module">
      // does NOT work
      console.debug(msg);
    </script>
  </body>
</html>
```

- *"msg" is undefined*
- *Variables defined in each <script> block are visible only within that block (local)*

```
// one.js
let msg = "Hello";
export { msg }
```

```
// two.js
let msg = "Hello World";
export { msg }
```

```
<html>
  <body>
    <!-- Because the two JS files are imported,
    the following two lines are NOT needed
    <script src="one.js" type="module"></script>
    <script src="two.js" type="module"></script> -->
    <script type="module">
      import { msg } from "./one.js";
      import { msg as msg2 } from "./two.js";
      console.debug(msg);
      console.debug(msg2);
    </script>
  </body>
</html>
```

ES6 Modules: exporting multiple items

Exporter astro.ts

```
// In astro.js (or astro.ts)
const planetNames = [
  "Mars",
  "Mercury",
  /* more data here */
];
function distToSun(planet: string): number {
  /* more code here */
}
export { planetNames, distToSun };
```

Just import what you need

```
// resolve to .js or .ts
import { planetNames } from "./astro";
console.log(planetNames[1]); // "Mercury"
```

```
import { distToSun } from "./astro";
console.log(distToSun("Venus"));
```

ES6 Modules: Default vs. Named Exports

Exporter astro.ts

```
// In astro.js (or astro.ts)
const planetNames = [
  "Mars",
  "Mercury",
  /* more data here */
];
function distToSun(planet: string): number {
  /* more code here */
}
export { planetNames };
export default distToSun;
```

- Default exports can be renamed however you like (at the time of import)
- Named exports must be imported verbatim

From Default

From Named Export

```
import computeDistance, { planetNames } from "./astro";
console.log(planetNames[1]); // "Mercury"
const marsToSun = computeDistance("Mars");
```

Choices of Modules

- CommonJS (2009): `require("module-name")` and `module.exports = { }`
 - Use by NodeJS
 - ES6 Modules (2015)
 - `import` and `export` (shown as examples in earlier slides)
 - AMD: Asynchronous Module Definition
 - RequireJS (supplement to AMD)
 - UMD: Universal Module Definition
 - enable apps to use CommonJS and AMD together
- } *Less popular*

CommonJS Modules (used by NodeJS)

astro.ts

```
const planetNames = [  
  "Mars",  
  "Mercury",  
  /* more data here */  
];  
function distToSun(planet: string): number {  
  /* more code here */  
}  
// Option #1  
module.exports = { planetNames, distToSun };
```

astro.ts

```
const planetNames = [  
  "Mars",  
  "Mercury",  
  /* more data here */  
];  
function distToSun(planet: string): number {  
  /* more code here */  
}  
// Option #2  
exports.planetNames = planetNames;  
exports.distToSun = distToSun;
```

```
const universe = require("../astro");  
console.log(universe.planetNames[1]); // "Mercury"  
const marsToSun = universe.distToSun("Mars");
```


AMD Modules

CommonJS

```
const planetNames = [
  "Mars",
  "Mercury",
  /* more data here */
];
function distTo(planet: string): number {
  /* more code here */
}
module.exports = {
  names: planetNames,
  distanceToSun: distTo,
};
```

CommonJS

```
const universe = require("../astro");
console.log(universe.names[1]); // "Mercury"
const marsToSun = universe.distanceToSun("Mars");
```

AMD

```
const planetNames = [
  "Mars",
  "Mercury",
  /* more data here */
];
function distTo(planet: string): number {
  /* more code here */
}

define(function () {
  return {
    names: planetNames,
    distanceToSun: distTo,
  };
});
```

AMD

```
require(["../astro"], function (universe) {
  console.log(universe.names[0]); // "Mars"
  const marsToSun = universe.distanceToSun("Mars");
});
```